

Road Damage Detection using YOLOv8

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1. Introduction

Road surface damages such as cracks and potholes affect both traffic safety and road quality. Manual inspection of roads is slow and subjective, making it difficult to scale. This project aims to build an automated road damage detection system using deep learning. Given a road image, the model identifies damaged regions, classifies the type of damage, and outputs bounding boxes with confidence scores. This work was developed as part of the Road Damage Detection 2022 (RDD2022) Crackathon.

2. Dataset

The RDD2022 dataset provided by the organizers was used in this project.

The dataset consists of real-world road images captured under diverse conditions. Data was divided into training, validation, and test sets.

Only the provided dataset was used, in accordance with competition rules.

Damage Categories

0 – Longitudinal Crack

1 – Transverse Crack

2 – Alligator Crack

3 – Other Corruption

4 – Pothole

3. Model Architecture

We employed YOLOv8 Small (YOLOv8s) for object detection.

- Framework: Ultralytics YOLOv8
- Pretrained on the COCO dataset

YOLOv8s was selected for its balance between accuracy and computational efficiency, making it suitable for detecting road damage in real-world scenarios.

4. Training Strategy

The model was trained using the labeled training data from the RDD2022 dataset.

- Epochs: 50
- Image size: 640 • 640
- Optimizer: Default YOLOv8 optimizer

Data Augmentation

Default YOLOv8

augmentations were applied, including mosaic augmentation, random horizontal flipping, HSV color augmentation, and scaling.

5. Validation Results

Model performance on the validation set was as follows:

- $mAP@50 \approx 0.52$
- $mAP@50-95 \approx 0.28$
- Precision ≈ 0.63
- Recall ≈ 0.50

Training and validation curves showed stable convergence without significant overfitting.

6. Implementation Details

Training was performed on Kaggle using GPU resources. Testing and submission preparation were carried out on a local machine.

The best model weights were selected based on validation performance.

7. Test Inference and Submission

Predictions were generated for the test dataset in YOLO format:

```
<class_id> <x_center>  
<y_center> <width> <height>  
<confidence>
```

One prediction file was generated per test image. Empty files were created for images with no detected damage. All prediction files were placed inside a predictions folder and compressed into submission.zip as required.

8. Conclusion

The proposed YOLOv8-based system successfully detects and classifies multiple types of road damage under real-world conditions. The approach adheres to competition guidelines and demonstrates the effectiveness of deep learning for automated road inspection.

